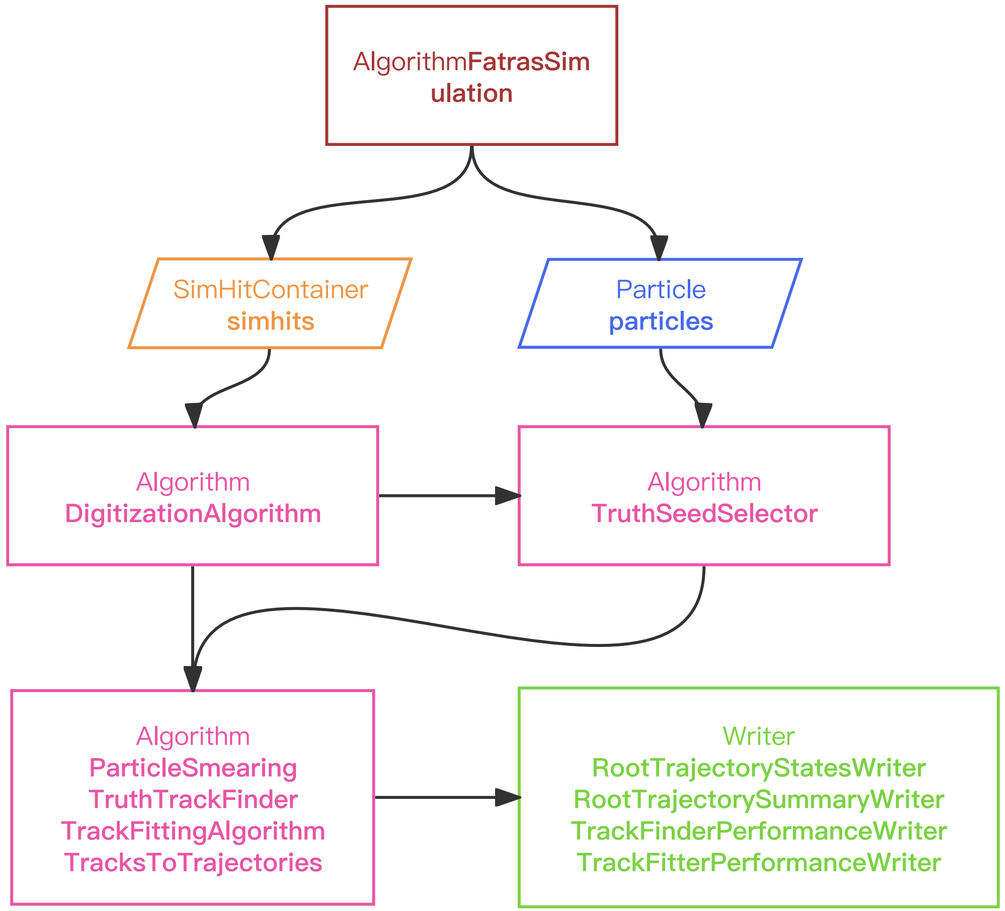
ACTS-DSS update

Xuliang Zhu, Yifan Zhu

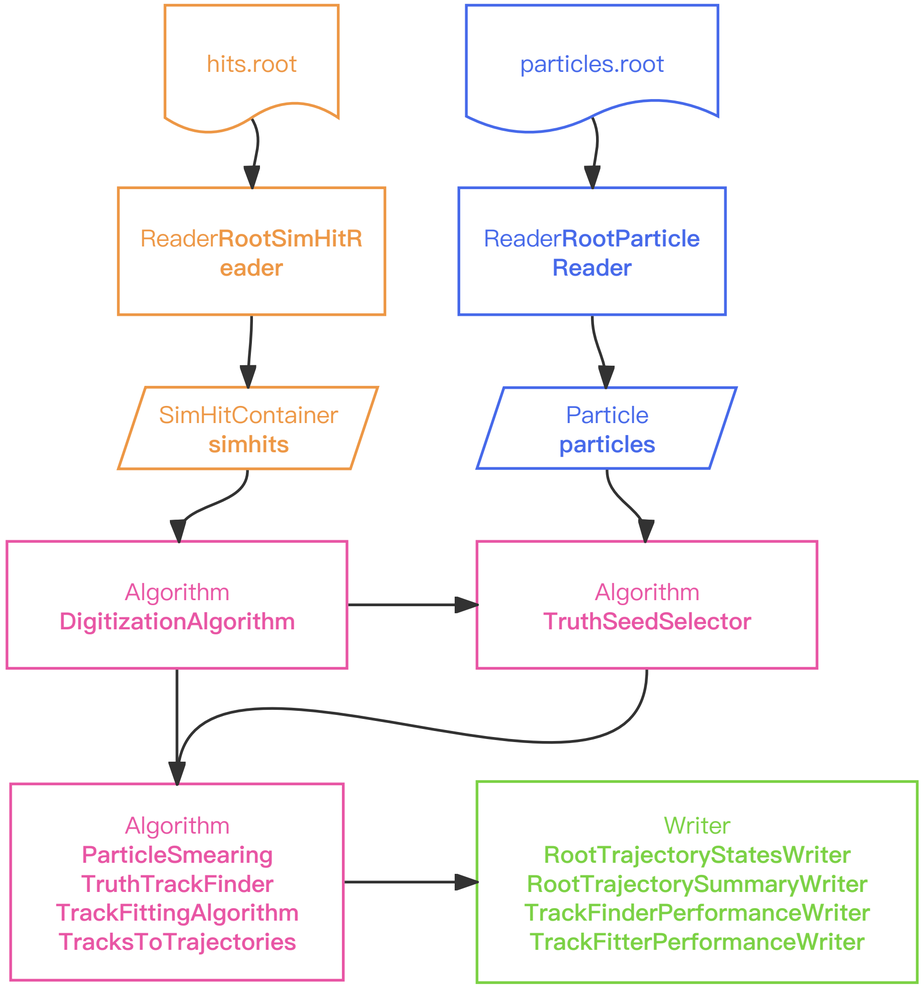
2023/10/17

ACTS Sequence

ACTS Example:
truth_tracking_telescope



For performance comparison:
DSimu -> DAna
DSimu -> hits.root, particles.root
-> *truth_tracking_telescope_reader*



hits.root

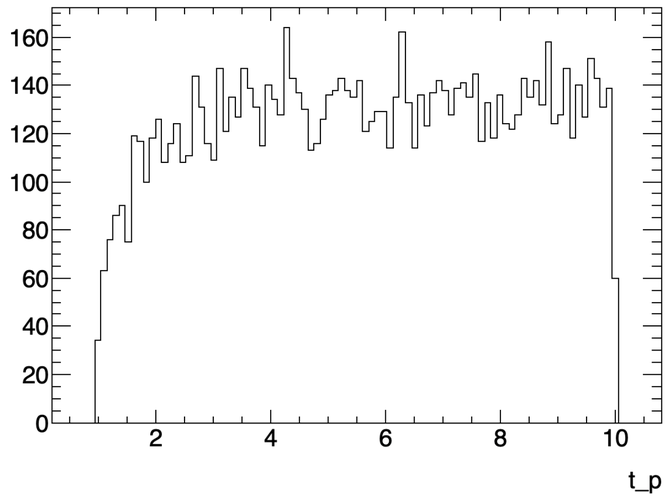
```
event_id;
geometry_id;
particle_id;
tx;
ty;
tz;
tt;
tpx;
tpy;
tpz;
te;
deltapx;
deltapy;
deltapz;
deltae;
index;
volume_id;
boundary_id;
layer_id;
approach_id;
sensitive_id;
```

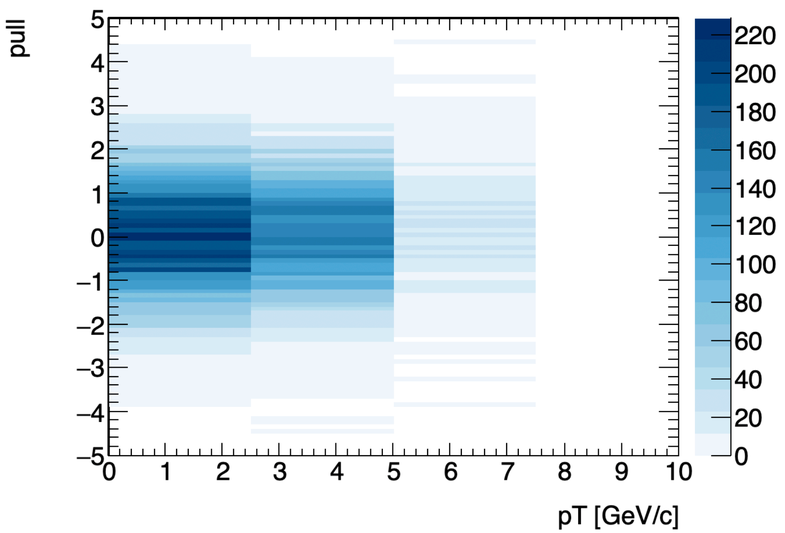
particles.root

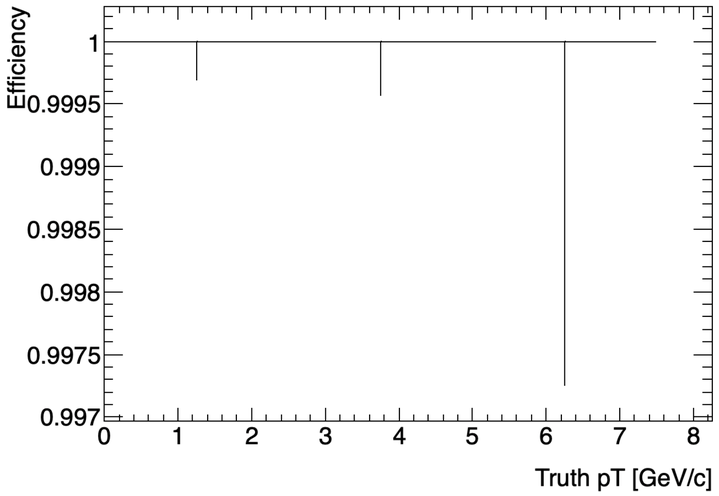
```
event_id;
*particle_id;
*particle_type;
*process;
*vx;
*vy;
*vz;
*vt;
*px;
*py;
*pz;
*m;
*q;
*eta;
*phi;
*pt;
*p;
*vertex_primary;
*vertex_secondary;
*particle;
*generation;
*sub_particle;
```

Fitter Performance

- Pull: the difference between the reconstructed and the true track parameter, divided by the estimated error on the reconstructed parameter.







Todo

- Compare reconstruction performance between ACTS & original DAna algorithm.